

colouring based on class labels, and comparison across different classifiers.

Visualise associations

Description: Visualises association rules discovered by mining algorithms, helping understand relationships and dependencies between data items.

Working: Rules are typically displayed as text strings like 'A & B => C', indicating that the presence of attributes A and B implies the presence of C.

Technical points: Supports various association rule mining algorithms (e.g., Apriori), allows setting minimum support and confidence thresholds, and offers interactive rule filtering.

WEKA and generative AI solutions

Before delving into generative AI, it is critical to comprehend the following WEKA characteristics.

WEKA's graphical user interface (GUI) is designed to be intuitive, catering to users of diverse machine learning proficiency levels. The utilisation of its drag-and-drop functionality streamlines the construction, assessment, and implementation of machine learning models.

It offers a comprehensive range of tools designed to facilitate data preprocessing, guaranteeing that the data is suitably prepared prior to its input into machine learning models. This consists of feature selection, normalisation, and the management of absent values.

WEKA comprises an extensive assortment of algorithms designed for machine learning, such as support vector machines, neural networks, clustering methods, and others. This diversity empowers users to explore and test out various methodologies for a wide range of duties.

In generative AI, new instances of synthetic data that resemble a given dataset are generated. In this context, WEKA has the capability to produce

novel patterns, images, and textual content. It can be implemented in the following specific generative AI scenarios.

GANs: Generative adversarial networks (GANs) are a widely used category of generative models in which a discriminator assesses the degree to which the generated data resembles actual data. WEKA, by utilising its extensive collection of algorithms to optimise the generator and discriminator components, can be employed to train and assess GANs.

Generation of text: The text mining functionalities of WEKA can be utilised to generate textual content. Users may train models to produce novel, coherent text by examining patterns in pre-existing textual data. This can prove to be especially beneficial in the domains of creative writing, content generation, and chatbot development.

Image generation: WEKA is a valuable instrument for generative image tasks due to its support for image processing and analysis. Users have the ability to train models to generate novel images by utilising algorithms like neural networks and decision trees, which determine the patterns to be extracted from pre-existing datasets.

Music customisation: It demonstrates a remarkable capacity for adaptability in the realm of music generation as well. Algorithms based on machine learning can be utilised by users to examine genres, chord progressions, and musical patterns. The model can then generate new musical compositions that draw inspiration from the patterns it has learned.

Here's how WEKA contributes to generative AI workflows.

▪ **Data preprocessing**

Cleaning and formatting: WEKA's filters can clean and format your data, handling tasks like missing value imputation, normalisation, and discretisation. This prepared data becomes suitable input for training

generative models.

Exploratory analysis: Visualisations like scatter plots, parallel coordinates, and cluster analysis help you understand data distribution, identify biases, and explore relationships, informing decisions about the generative model's design and training.

▪ **Feature engineering (indirectly)**

While WEKA doesn't offer dedicated feature engineering tools, you can use it to analyse derived attributes and feature importance, providing insights for creating new features potentially useful for generating realistic data.

▪ **Evaluation**

After training your generative model (using external libraries like TensorFlow or PyTorch), you can import the generated data into WEKA for evaluation. Classification or clustering tools assess the quality of generated data compared to real data, using metrics like accuracy, precision, recall, or similarity measures.

▪ **Visualisation of generated data**
Visualisations in WEKA help explore and understand the generated data itself. It can also analyse distributions, relationships, and clusters to identify potential issues, biases, or deviations from real data.

WEKA acts as a supporting tool in generative AI, not a primary implementation platform. Core model training and generation rely on external libraries or frameworks specifically designed for generative tasks. WEKA's strength lies in data analysis and preparation, making it valuable for building a robust workflow around generative AI models.

Strengths of WEKA

▪ **Wide range of algorithms:** It boasts a diverse collection of algorithms covering supervised learning (classification, regression), unsupervised learning (clustering, association rule mining), and evaluation techniques. This versatility allows you to tackle